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Tropical cyclone risk to global mangrove ecosystems: potential future regional shifts

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Tropical cyclones pose a major risk to mangroves, yet the evolution of risk in response to altered cyclone activity under climate change remains largely unexplored. Analyzing 1886 tropical cyclones (1980–2020), we found that mangrove ecosystems were most prone to major cyclones (Category 3–5), and that intense cyclones (Category 4–5) contributed 97% of the global risk of damage. Risk hotspots were found in mangroves bordering the Gulf of Mexico/Caribbean Sea, the South Indian Ocean, and the Northwest Pacific. With warming of 2°C, the risk changes were projected to be modest (+3%) at the global scale but substantial and divergent at the regional level. Risk in North America (including Central America) and Oceania was predicted to increase and decrease by more than 10%, respectively. Our results identify regions where cyclone-activity-induced ecosystem changes may be expected and where mangrove management needs to prioritize cyclone resilience.

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Mangroves are important coastal ecosystems that provide a diversity of economic, social, and ecological services – including food provision, water purification, biodiversity conservation, storm protection, and carbon sequestration – valued at over US$20,000 ha^{-1} yr^{-1} (Temmerman *et al.* 2013; Del Valle *et al.* 2020; Su *et al.* 2021). Tropical cyclones (TCs) are the most destructive natural disturbance faced by mangroves, causing 45% of their naturally induced mortality, and in some regions (such as North America and Oceania) are a primary driver of mangrove loss (Sippo *et al.* 2018; Goldberg *et al.* 2020). In addition, the damage to individual trees is mechanistically linked to mangrove ecosystem structure and function (Simard *et al.* 2019; Lin *et al.* 2020). Therefore, a more thorough understanding of the TC-associated risk of damage is imperative for improving mangrove management.

A TC can damage mangroves directly through physical forces such as wind and storm surges and indirectly by inducing hydrological and environmental changes such as ponding and hypersalinization. At the community level, such damage usually manifests as defoliation, canopy cover reduction, and tree mortality (Hogan *et al.* 2020; Krauss and Osland 2020). The level of damage at a given site is influenced by many factors, including the storm's attributes (such as wind speed and accumulated rainfall) and the site's relative location to the storm track (such as distance and quadrant to the track) (Svejkovsky *et al.* 2020; Taillie *et al.* 2020). Damage levels may also be influenced by the characteristics of the mangrove ecosystems (such as species composition, canopy height, and tree density) (Armitage *et al.* 2019; Walcker *et al.* 2019; Pennings *et al.* 2021) and by the features of the coastal environment (such as elevation and historical cyclone frequency) (Taillie *et al.* 2020; Lagomasino *et al.* 2021). Furthermore, the damage could be aggravated by prior natural and anthropogenic disturbances (such as chilling and anthropogenic hydrological modification) (Zhang *et al.* 2016; Walcker *et al.* 2019).

Advances in satellite-based Earth observations have recently enabled large-scale studies to sample multiple (tens of) storms, demonstrating the effectiveness of predicting TC damage to mangroves at the regional and global scale with storm intensity (that is, [local] wind speed and accumulated rainfall [Taillie *et al.* 2020], and wind speed and distance to track [Peereman *et al.* 2021]). Climate change is projected to affect TC activity, with global TC intensity and frequency expected to increase and decrease, respectively (eg Knutson *et al.* 2020). Crucial questions, therefore, arise. How may the risk of damage change as TC activity changes under future climate conditions? And more importantly, how will the risk evolve regionally? Addressing these questions is critical for developing mangrove management strategies under climate change.

Here, we tackled these questions by conducting a global study based on storm records and continuous satellite-based Earth observations over the past four decades. We sampled 1886 Category 1–5 TCs (1980–2020) to systematically quantify the damage (measured by remote-sensed vegetation index) and risk (combining damage and frequency) to mangrove ecosystems induced by TCs of different intensities (ie categories). We then estimated the risk evolution at global and regional scales based on projected changes in TC activity.

Methods

Cyclone records

As identified from the International Best Track Archive for Climate Stewardship (IBTrACS V4) using the

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Saffir–Simpson wind scale (SSWS), a total of 1886 TCs (Category 1 to 5) occurred around the world between 1980 and 2020 ([Knapp *et al.* 2018; NOAA 2021]; Figure 1; see WebPanel 1 for more details). The SSWS was determined based on the TC's lifetime-maximum wind speed, so as to align with the TC activity change projections. For each category of TC, a map of annual frequency at 1° resolution was created using the recorded cyclone track points. The regional-scale TC activity changes projected with a 2°C increase in global mean temperature were adapted from a systematic assessment conducted by Knutson *et al.* (2020), which incorporated studies reviewed in the Intergovernmental Panel on Climate Change Fifth Report and more recent work. TC damage to mangroves was evaluated with TCs that occurred after 2001.

Mangrove extent

The global mangrove extent was delineated based on the Global Mangrove Forests Distribution (GMFD) dataset (Figure 1; Giri *et al.* 2011). The GMFD dataset was chosen because it was specifically derived from Landsat data and coincided with the start of our study period for assessing TC damage to mangroves (ie 2000). We minimized the impact of changes in mangrove extent on our results by computing the pre-cyclone baseline conditions for each TC event and then identifying changes relative to the baselines (see WebPanel 1 for more details). TC frequencies within mangrove ecosystems of each of the 120 countries, dependent territories, and special areas of geographical interest defined in the GMFD dataset (hereafter, "countries") were calculated by intersecting the TC frequency maps with the country's mangrove extent (with a 1° buffer zone, to account for all potential TC damage).

Tropical cyclone damage to mangroves

For each TC event, the conditions of mangroves within 100 km on either side of the TC's track were monitored for 6 years (1 year before the TC to 5 years after the TC) using Moderate Resolution Imaging Spectroradiometer (MODIS) Terra and Aqua Enhanced Vegetation Index (EVI) composites (WebPanel 1; Didan *et al.* 2015). We assessed mangrove conditions using five variables: EVI, areal extent, number of patches, average patch size, and size of the largest patch. Measurements from the 12 months before a TC event (eg July 2019–June 2020, if the TC occurred in July 2020) were considered to be baseline conditions, and all measurements were normalized against their baselines.

The normalized measurements were then compiled for each category of TC, grouped into 3-month time steps, and tested using one-way analysis of variance (ANOVA) and Tukey post-hoc tests (significance at $P < 0.05$; WebPanel 1). A TC Damage Index (TCDI) was calculated by aggregating significant decreases in normalized EVI ($\widehat{\mathrm{EVI}}$) over time (t) and distance (d) for each category (cat) of TC:

$$\mathrm{TCDI}_{\mathrm{cat}} = \sum^{t,d} 1 - \widehat{\mathrm{EVI}}_{\mathrm{cat}} \qquad \text{(Equation 1).}$$

The uncertainty of the TCDI was calculated by pooling the variance of the EVI measurements. The TC Risk Index (TCRI) was computed by multiplying the TCDI and the frequency (Freq) of TC of that category:

$$\mathrm{TCRI}_{\mathrm{cat}} = \mathrm{TCDI}_{\mathrm{cat}} \times \mathrm{Freq}_{\mathrm{cat}} \qquad \text{(Equation 2).}$$

The uncertainty of the TCRI was calculated by propagating the uncertainties of the TCRI and the TC frequency.

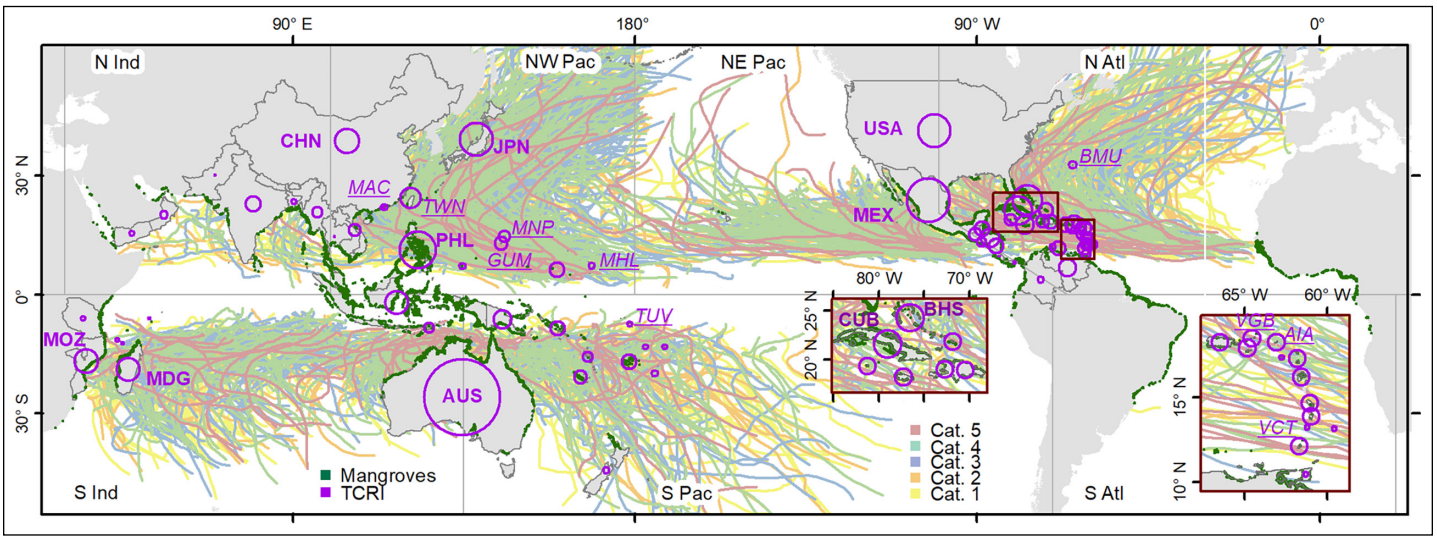

Figure 1. Tropical cyclones (TCs) and the risk they pose to global mangrove ecosystems. TC categories were determined based on the Saffir–Simpson wind scale and the lifetime-maximum wind speed of the storms. Areas within the purple circles are proportional to the Tropical Cyclone Risk Index (TCRI) of different countries. The abbreviated names of the ten countries with the highest total risk (Australia, Bahamas, China, Cuba, Japan, Madagascar, Mexico, Mozambique, Philippines, US) and with the highest risk per unit area of mangroves (Anguilla, Bermuda, British Virgin Islands, Guam, Macau, Marshall Islands, Northern Mariana Islands, Saint Vincent and the Grenadines, Taiwan, Tuvalu) appear in bold and in italics, respectively.

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Results and discussion

Damage extent and ratio

Damage to mangrove systems was detected only as a result of major TCs (Category 3 or higher), and primarily from intense TCs (Category 4 and 5) (Figure 2; WebFigure 1). Damage was exhibited as decreases in EVI and areal extent and as an increase in mangrove canopy fragmentation (ie the average patch size and the largest patch size decreased, and the number of patches increased). Using EVI as the measure, the damage from a Category 3, 4, and 5 TC (i) led to (on average) a 5%, 10%, and 20% reduction, respectively, in pre-cyclone baseline values; (ii) was detected up to 40 km, 50 km, and 50 km, respectively, away from the cyclone track; and (iii) recovered in (on average) 3 months, 1 year, and 2 years, respectively (Figure 2; WebFigure 2). Integrating the damage over space and time, the ratio of the overall damage of a Category 5, 4, and 3 TC was estimated to be 29:13:1 (inferred by TCDI; WebFigure 3).

Our results suggest that only if a storm develops into a major TC (Category 3–5) will it be likely to cause substantial damage to mangrove ecosystems. This is in general agreement with existing observations that major TCs almost always cause extremely visible effects on mangroves (Krauss and Osland 2020). We recognize that wind speed at landfall – along with other factors influencing local wind speed, such as distance and quadrant to the track – is a more direct predictor of damage at a given site (eg Svejkovsky *et al.* 2020; Taillie *et al.* 2020). To interpret our findings under that perspective, we examined the wind speed at landfall for all Category 1–5 TCs over the past four decades. We found that the median values of maximum wind speed at landfall for Category 1–2 TCs were between 50 and 70 km hr^{-1} and those for Category 3–5 TCs were between 75 and 100 km hr^{-1} (WebPanel 1). Therefore, our results also agree with a previous study (based on the 2017 North Atlantic hurricane season) showing that the probability of damage is very low (~0.1) when local maximum wind speed is less than 75 km hr^{-1} (Taillie *et al.* 2020).

The damage magnitudes and durations revealed by our analysis are within the lower range of values previously reported (ie magnitudes of 3–100% decrease in canopy cover and duration of 3 months to over 10 years, if not permanent; WebPanel 2; note that TC categories reported in previous studies were re-categorized based on lifetime-maximum wind speed to align with our analysis). The values reported here are on the conservative side, partially because (i) we focused on the average damage in 10-km-wide zones, whereas the greatest damage (over 50% decrease in vegetation index) mostly occurred at scales less than 1 km (eg Svejkovsky *et al.* 2020) and (ii) previous in situ studies were likely to focus on sites with the most damage (eg Asbridge *et al.* 2018). The small magnitude and short recovery time of TC-induced damage to mangroves at the landscape scale conforms to the theory that mangrove systems are adapted to TC disturbance (Krauss and Osland 2020). On the other hand, with respect to the maximum distances from cyclone tracks at which damage occurred, the values that we observed here were higher than most of the values reported in the literature, largely because the majority of previous studies were not designed to detect the extent of damage.

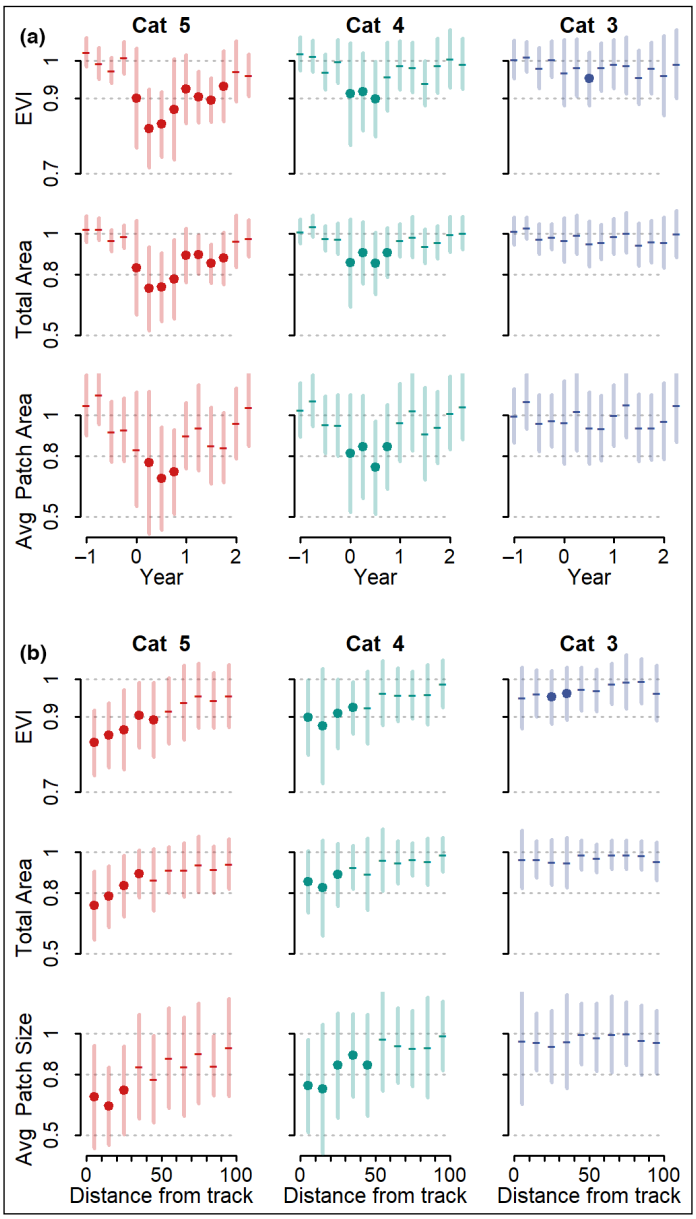

Figure 2. The temporal and spatial characteristics of the damage of major tropical cyclones to mangrove ecosystems. Means are shown with dashes (baseline values) and solid circles (values differ significantly from the baselines at $P < 0.05$), and standard deviations are shown in bars. (a) Mangrove conditions before and after TCs (at the 0–10 km zone for Category 4–5 TCs and the 20–30 km zone for Category 3 TCs). (b) Mangrove condition changes over distance (6 months following the cyclone).

The regain in EVI that occurred after cyclones may have reflected mangrove canopy recovery through re-foliation, re-sprouting, and sapling and seedling growth (Krauss and Osland 2020), and possibly also the growth of understory

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herbaceous plants (Svejkovsky *et al.* 2020). The fact that the mangrove's EVI returned to pre-cyclone levels, as shown in our results, highlights mangrove recovery, as mangroves have higher EVI than herbaceous plants. The 2-year recovery period for the areas that experienced the greatest damage (ie within 20 km of the track of Category 5 TCs; WebFigure 2) is commensurate with the growth of mangrove seedlings to juveniles (eg López-Hoffman *et al.* 2007), which might have been the main recovery pathway in those areas.

Intensified and delayed damage

In addition to the general patterns of recovery from disturbance, the temporal and spatial characteristics of the damage also exhibited some interesting features that might shed light on damage mechanisms. For example, intensified damage was found within the 10–20 km and 40–50 km zones for Category 4–5 TCs, and damage from Category 3 TCs was detected only within the 20–30 km and 30–40 km zones (Figure 2; WebFigure 1). These patterns may reflect the distribution of the TC's physical forces (ie the cyclone's primary and secondary eyewalls that are usually located at 15 km and 50 km from the storm center, respectively; Houze *et al.* 2006). Another possible explanation for this phenomenon is that many TCs may have passed the mangroves on their seaward side (without making landfall), and these patterns reflect the high resistance of mangroves at the shoreline due to shorter canopy heights and protection from strong winds as a result of being submerged by storm surges (Krauss and Osland 2020), an effect that becomes less relevant as winds become more powerful (Category 4–5).

Another notable finding is that damage was only detected 6–9 months after Category 3 TCs, and that the most severe damage was observed 6–9 months after Category 4 TCs and 3–6 months after Category 5 TCs. Recent studies have reported delayed tree mortality; for instance, tree mortality in Florida following Hurricane Irma in 2017 occurred 9 months after the storm, with one of the largest recorded dieback events (over 10,000 ha) occurring a full 15 months afterward (Radabaugh *et al.* 2020; Lagomasino *et al.* 2021). The delayed mortality and dieback in the months after Irma were primarily attributed to storm-induced hydrological changes (ie mud deposition and ponding), which can lead to stresses like oxygen limitation, phytotoxin accumulation, and hypersalinization (Radabaugh *et al.* 2020; Lagomasino *et al.* 2021). Defoliation that occurred during storms can increase mangrove susceptibility to these stresses, given that leaves are essential for mangrove salt regulation (Asbridge *et al.* 2018). Our analysis detected delayed impacts on a global scale, indicating that damage from storm-induced hydrological changes may be far more substantial than previously recognized.

Risk distribution and projected changes

Integrating with their frequency, at the global level, intense TCs composed 97% of the risk of damage (inferred by TCRI; WebFigure 3). At the country level, the risk was widely distributed across the globe (Figure 1). Hotspots of risk were found in mangroves bordering the Gulf/Caribbean, the South Indian Ocean, and the Northwest Pacific. In addition to the well-recognized high risk in the Gulf/Caribbean region and Oceania (Goldberg *et al.* 2020; Krauss and Osland 2020), our results highlight the risk within the Northwest Pacific, revealing that the region contains three of the ten countries with the highest total risk (China, Japan, and the Philippines) and four of the ten countries with the highest risk per unit area of mangroves (Guam, the Marshall Islands, the Northern Mariana Islands, and Taiwan).

With a 2°C increase in global mean temperature, we estimated that future global TC frequency (within mangrove habitats) would marginally decrease (–2%); however, with an increase in the frequency of intense TCs, future global risk might slightly increase (+3%) (WebFigure 3). At the regional level, on the other hand, changes could be substantial. The risk to most mangroves around the Northeast Pacific, North Atlantic, and North and South Indian Ocean was predicted to increase, while the risk to mangroves bordering the Northwest Pacific and Southwest Pacific was predicted to decrease (Figure 3). Notably, the risk to mangroves in North America (including Central America) and the Caribbean was projected to increase by over 10%, and the risk to mangroves in Oceania was projected to decrease by over 10%.

An increase in global storm risk to mangroves under climate change is often suggested in the scientific literature, given the projected increase in future storm intensity (Sippo *et al.* 2018; Krauss and Osland 2020). Our results suggest that the increase in global risk might be modest, and that major shifts could happen at the regional level. The changes in risk are mostly driven by projected changes in the activity of intense TCs (Category 4–5), which in turn is a combined result of the projected changes in TC genesis and intensity (ie the future frequency of intense TCs is determined by future total TC frequency and the proportion of future TCs that will reach Category 4–5 intensity). The projected increase in TC intensity is probably driven primarily by projected increases in sea-surface temperature (SST), while the physical mechanisms responsible for the projected decreases in TC frequency are less clear and may involve factors including SST, atmospheric moisture and temperature profiles, and upward mass flux (Knutson *et al.* 2020; Chand *et al.* 2022). Consequently, regional variation in SST is likely to be a key driver of the divergent changes in future intense TC activity and therefore storm risk to mangroves.

It should be noted that the projections of TC intensity have greater agreement among existing studies than do the projections of TC frequency, and that projections are more robust at the global scale than for individual basins (Knutson *et al.* 2020). To account for uncertainty in projections of TC activity, we used values from a synthesized assessment involving numerous studies (eg Knutson *et al.* 2020). We also incorporated uncertainty of the TC activity projections into the uncertainty of our TC risk projections (WebFigure 4). It should also be

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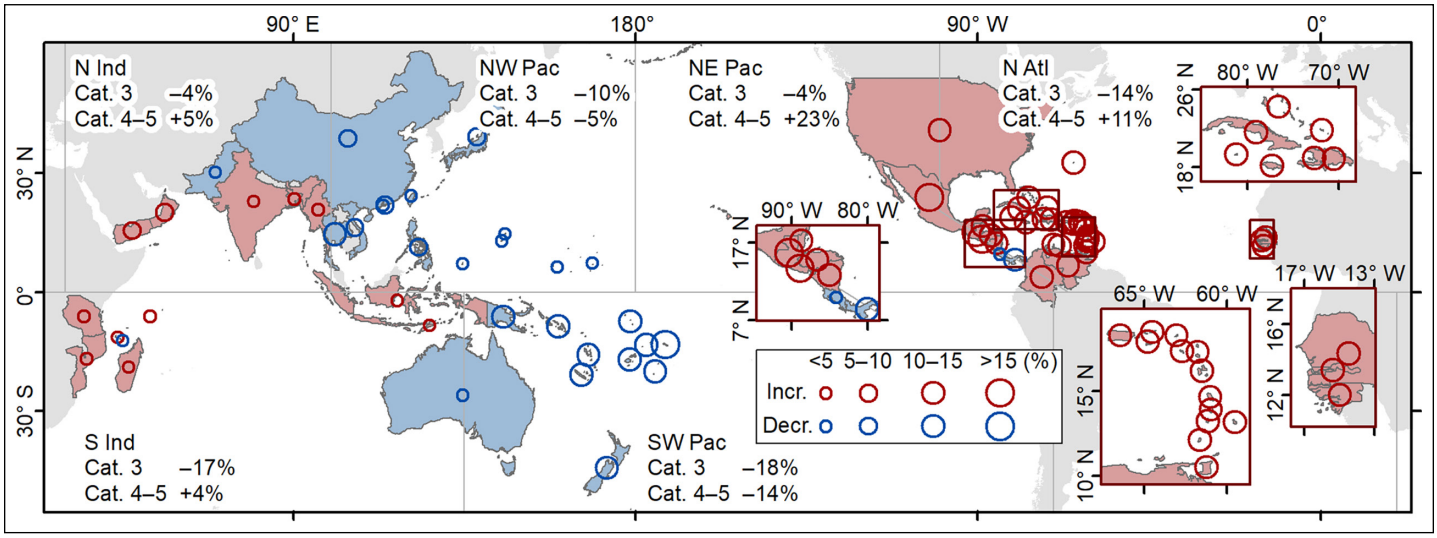

Figure 3. Risk changes under conditions of 2°C warming. Countries facing increased risk are shown in red and those facing decreased risk are shown in blue. Changes in the annual frequency of major TCs (Category 3 or higher) in different ocean basins are also noted on the map.

noted that our analysis did not account for (i) changes in subregional TC activity (Bloemendaal *et al.* 2022), which are likely to lead to more complex spatial patterns of shifts in risk, or (ii) projected increases in sea level and storm-induced surges, precipitation, and flooding, which are likely to amplify TC damage and thus risk (Woodruff *et al.* 2013).

Implications

Our TC risk assessment highlights regions where natural resource managers should prioritize actions to enhance mangrove resilience. For example, in Asia, fast-growing but storm-susceptible *Rhizophora* species are often used in mangrove restoration projects to facilitate rapid expansion of mangrove area at relatively low cost (Lee *et al.* 2019). Our study identifies areas that face high TC risk, where the use of more storm-resilient species would therefore be required. In addition, our results projected major shifts in TC risk to mangrove ecosystems under climate change, signifying potential changes in community structure and thus ecosystem functions, such as carbon sequestration and biodiversity conservation. For instance, mangroves that face increasing TC risk are likely to have reduced average tree height and favor the establishment of shade-intolerant species (Lin *et al.* 2020). Such effects may also be manifest in mangrove encroachment at high latitudes in response to warming winter temperatures, with the new mangrove communities possibly differing greatly from the old mangrove forests in regions expected to experience dramatic changes in TC risk.

The observed delay in damage (an average lag of 6 months) also has major research and management implications. For example, it suggests that to fully capture the damage incurred, assessments must be conducted in the months following a storm and not simply immediately afterward. It also highlights the need for further investigation into the potentially profound impacts of cyclone-induced environmental change. Studies can compare the damage progression (or recovery) in sites subjected to direct (ie via the physical forces of storm wind and surge) or indirect (ie via cyclone-induced hydrological changes such as ponding) damage. Research may also compare the positive (eg fertilization; Castaneda-Moya *et al.* 2020) and negative (eg smothering; Radabaugh *et al.* 2020; Lagomasino *et al.* 2021) effects of storm-induced sedimentation in those sites. Furthermore, human activities, such as road and levee construction, may have modified the hydrologic conditions of mangrove habitats and increased their susceptibility to storm-induced changes (Walcker *et al.* 2019). A better understanding of the impacts of storm-induced hydrological changes would provide additional guidance to these activities. This knowledge may also be critical to the management of coastal ecosystems more broadly, as these ecosystems often exhibit similar responses to disturbance (Patrick *et al.* 2022).

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Data Availability Statement

The code for acquiring and analyzing data, as well as the data underlying the figures, can be accessed on the project GitHub page at: https://github.com/github-moyu/CHES.git.

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